Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 29

Other Starting Points, Inequalities, and Divisibility

Covers Chartrand & Zhang, Section 4.2

Status of this set

Ungraded practice. Solutions released with it. Attempt everything first.

Part I: How to Practice This Material

Three things that change from Lecture 28

1. **The base step may not be at 1.** Test small values first; start where the claim actually becomes true.
2. **Inequalities chain.** Start from one side of $P(k+1)$, bring in the hypothesis, and justify every link until you reach the other side.
3. **Divisibility needs the k case to appear.** Rewrite the $k+1$ expression so that the k expression shows up inside it — then substitute.

Three mistakes to expect

1. **Starting at $n=1$ for a claim false at $n=1$.**
2. **An unjustified link** in an inequality chain.
3. **Dividing by something that might be zero** or flipping an inequality by multiplying by a negative.

Part II: Practice Problems

Section A — Inequalities

1. Prove that $3^n \geq 2n+1$ for every positive integer n.

2. Test $n^2 > 2n + 1$ for $n = 1, 2, 3, 4$. Identify where it starts to hold, and prove it from that point on.

Section B — Divisibility

3. Prove that 5 divides $6^n - 1$ for every positive integer n.

4. Prove that 6 divides $n^3 - n$ for every positive integer n.

Section C — Where the base goes

5. A student proves $n! > 3^n$ "for every integer $n \geq 1$" by checking only the inductive step. Test the claim for small n, find where it actually begins, and identify exactly what the student's proof would have missed.

Section D — Writing

6. In five to eight complete sentences, explain how the proof that $2^n > n^2$ for $n \geq 5$ settles Practice Set 1, Problem 15. Say what you could and could not conclude in Week 1, and what induction adds.